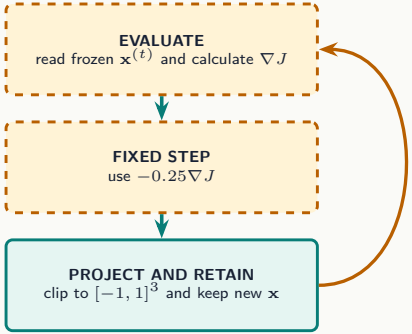
Plain projected gradient descent in one loop

SYNC BEAT 1 OF 7 · STATE NOW
TEACHING CONTROL: keep the teaching objective and starting point used in the L-BFGS deck, but replace its optimizer. Retain $\mathbf{x}^{(t)}$.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

FREEZE $\mathbf{x}^{(t)}$ ⟶ **CALCULATE ∇J** ⟶ **FIXED STEP $\eta = 0.25$** ⟶ **PROJECT AND REPEAT**



Objective function in this teaching control

$$J(\mathbf{x}) = -\frac{1}{2}\sum_{i=1}^{3} \ell_i x_i + 2 \sum_{c\in\{A,B\}} \left(1 - \prod_{j\in c} x_j\right)^2$$

One update

$$\mathbf{x}^{(t+1)} = \Pi_{[-1,1]^3}\left(\mathbf{x}^{(t)} - 0.25\nabla J(\mathbf{x}^{(t)})\right)$$

No $\mathbf{H}_t$, no $(\mathbf{s}, \mathbf{y})$ pairs, and no line search.
Constructed fixed-step control; not the retained native projected-GD endpoint.

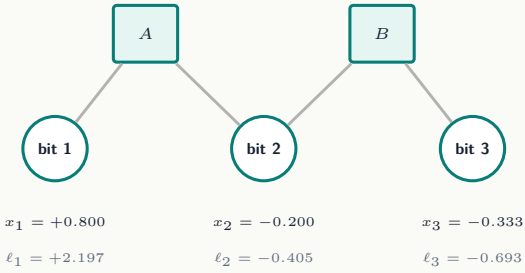
Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

Initialize once: channel evidence and one scalar per bit

SYNC BEAT 2 OF 7 · STATE NOW
INITIAL VALUES: store fixed $\boldsymbol{\ell} = -\mathbf{L}$ and initialize $\mathbf{x}^{(0)} = \tanh(\boldsymbol{\ell}/2)$. There is no optimizer history.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

PLACE FIXED $\boldsymbol{\ell}$ ⟶ INITIALIZE $\mathbf{x}^{(0)}$ ⟶ KEEP ONE SCALAR PER BIT



$$\boldsymbol{\ell} = -\mathbf{L} = (+2.197, -0.405, -0.693)$$

$$\mathbf{x}^{(0)} = \tanh(\boldsymbol{\ell}/2) = (+0.800, -0.200, -0.333)$$

PERSISTENT NUMERICAL VALUES

fixed $\boldsymbol{\ell}[3]$	3
current $\mathbf{x}^{(0)}[3]$	3
optimizer history	none
persistent total	**6**

Here $\ell_i > 0$ and $x_i > 0$ favour bit 0; the BP deck uses the opposite-sign channel LLR $L_i = -\ell_i$.

Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

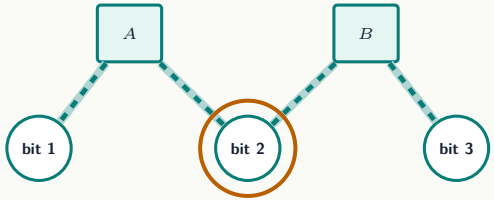
Frozen-state read: both checks calculate their residuals

SYNC BEAT 3 OF 7 · STATE NOW
LOCKED READ: checks A and B read one frozen $\mathbf{x}^{(0)}$. The residuals are temporary; no coordinate changes during this slide.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

LOCK $\mathbf{x}^{(0)}$ $\longrightarrow$ **A READS x_1, x_2** + **B READS x_2, x_3** $\longrightarrow$ **CALCULATE r_A, r_B**

$r_A = 1.160$ $r_B = 0.933333$



$$r_c = 1 - \prod_{j \in c} x_j$$

$$r_A = 1 - (0.800)(-0.200) = 1.160,$$
$$r_B = 1 - (-0.200)(-0.333333) = 0.933333.$$

Showing A before B is presentation order. Both calculations use the frozen vector.

The dashed lines show mathematical dependence on the coordinates; they do not show travelling messages.

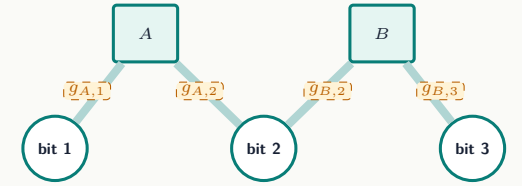
Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

Each check contributes terms to the gradient

SYNC BEAT 4 OF 7 · STATE NOW
ONE FROZEN SNAPSHOT: all four $g_{c,i}$ terms use $\mathbf{x}^{(0)}$. They are indexed by check and coordinate only while ∇J is calculated.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

READ FROZEN $\mathbf{x}^{(0)}$ AND r_c ⟶ CALCULATE FOUR $g_{c,i}$ TERMS ⟶ KEEP $\mathbf{x}^{(0)}$ FIXED



$x_1 = +0.800$ $x_2 = -0.200$ $x_3 = -0.333$

$$g_{c,i} = -4r_c \prod_{\substack{j \in c \\ j \neq i}} x_j$$

check A
$g_{A,1} = +0.928,$
$g_{A,2} = -3.712$

check B
$g_{B,2} = +1.244,$
$g_{B,3} = +0.747$

The four terms are temporary. They are combined by bit coordinate on the next slide.

The values contribute to the gradient of the objective function; they are not directed BP messages.

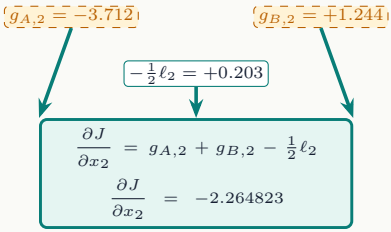
Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

One sum per coordinate: GD has no recipient exclusion

SYNC BEAT 5 OF 7 · STATE NOW
AGGREGATE ONLY: $\mathbf{x}^{(0)}$ remains frozen while every applicable term is added into one gradient coordinate. No outgoing check address is created.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

GROUP TERMS BY BIT INDEX ⟶ **SUM EVERY APPLICABLE TERM** ⟶ **FORM ONE $\partial J/\partial x_i$ PER BIT**



Complete gradient at the frozen point

$$\nabla J(\mathbf{x}^{(0)}) = \begin{bmatrix} g_{A,1} - \frac{1}{2}\ell_1 \\ g_{A,2} + g_{B,2} - \frac{1}{2}\ell_2 \\ g_{B,3} - \frac{1}{2}\ell_3 \end{bmatrix} = \begin{bmatrix} -0.170612 \\ -2.264823 \\ +1.093240 \end{bmatrix}$$

WHY THIS DIFFERS FROM BP SLIDE 5
Bit 2 receives terms associated with both checks A and B. GD adds them into one coordinate $\nabla_2 J$. It does not remove a recipient's contribution because it is not preparing an addressed message.

All applicable terms become one coordinate update; only the resulting x_2 can cross the next iteration boundary.

Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

Move once: the next evaluation still has one bit-2 scalar

SYNC BEAT 6 OF 7 · STATE NOW
START: $\mathbf{x}^{(0)}$ and $\nabla J(\mathbf{x}^{(0)})$. **END:** project the fixed step and retain $\mathbf{x}^{(1)}$; the temporary terms are discarded.

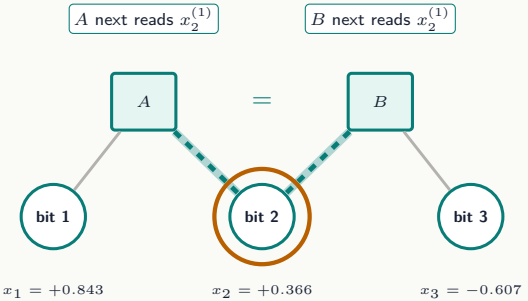
PERSISTENT STATE VALUES
fixed ℓ: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

NEGATIVE GRADIENT ⟶ **FIXED STEP** 0.25 ⟶ **PROJECT TO** $[-1,1]^3$ ⟶ **RETAIN** $\mathbf{x}^{(1)}$

$$-\nabla J(\mathbf{x}^{(0)}) = (+0.170612, +2.264823, -1.093240)$$

$$\begin{aligned}\mathbf{z}^{(0)} &= \mathbf{x}^{(0)} - 0.25\nabla J(\mathbf{x}^{(0)}) \\ &= (+0.842653, +0.366206, -0.606643), \\ \mathbf{x}^{(1)} &= \Pi_{[-1,1]^3}(\mathbf{z}^{(0)}) = \mathbf{z}^{(0)}.\end{aligned}$$

$$J(\mathbf{x}^{(0)}) = 3.398461 \longrightarrow J(\mathbf{x}^{(1)}) = 2.881689$$
No coordinate is clipped on this step.



Both checks next read the one retained $x_2^{(1)}$. This first point matches the teaching L-BFGS path because its empty-history direction is $-\nabla J$ and its line search accepts $\alpha = 0.25$; later directions need not agree.

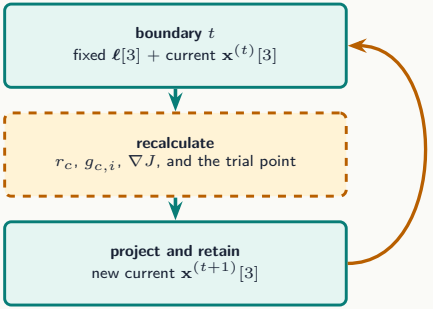
Persistent count includes fixed ℓ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity

What crosses a plain projected-GD iteration boundary

SYNC BEAT 7 OF 7 · STATE NOW
AFTER THE STEP: fixed $\boldsymbol{\ell}$ and current $\mathbf{x}^{(1)}$ remain. Residuals, $g_{c,i}$ terms, ∇J, and the trial point are recalculated.

PERSISTENT STATE VALUES
fixed $\boldsymbol{\ell}$: 3 current $\mathbf{x}$: 3
optimizer history: none **total: 6**

KEEP FIXED ℓ AND CURRENT $\mathbf{x}$ ⟶ RECALCULATE TEMPORARY TERMS ⟶ PROJECT THE NEXT FIXED STEP



PERSISTENT AFTER THE FIRST STEP

fixed channel evidence $\boldsymbol{\ell}$	3 values
current relaxed vector $\mathbf{x}^{(1)}$	3 values
optimizer history	none
persistent total	**6 values**

RECALCULATED, NOT RETAINED
r_A, r_B; four $g_{c,i}$ terms; ∇J; negative-gradient step; raw proposal and reusable projection workspace.
A concrete implementation needs reusable working storage. Its peak depends on the chosen evaluation schedule, so six is a persistent-state count, not a universal memory bound.

The objective decrease on this toy update demonstrates the calculation only; it does not establish decoding performance.

Mechanism contrast
BP retains separately addressed edge messages. This projected-GD control retains one scalar x_i per bit.

Persistent count includes fixed $\boldsymbol{\ell}$ and current $\mathbf{x}$; graph data and temporary arithmetic are excluded; synchronized slides do not imply equal arithmetic or complexity